

Benchmarking Progress Update

- Expanded from single-dataset to multi-dataset evaluation
 - RQ1: Is performance due to KG circularity?
 - RQ2: What factors drive AA-KEA performance?

Datasets Overview

- **Semantic-KG (Codex):** KG-perturbed baseline (92 pairs, ~69 words)
- **MRPC:** Human news paraphrases (400 pairs, ~20 words)
- **PAWS-Wiki:** Adversarial word-scrambled (400 pairs, ~20 words)
- **Semantic-KG Combined:** 4 domains (400 pairs)
- **Wikipedia Entity-Swap:** NLP-perturbed, anti-circularity probe (399 pairs), constructed using four NLP-based perturbations (node replacement, node deletion, edge deletion, edge replacement via WordNet antonyms) that mirror Semantic-KG's taxonomy but use no KG at any stage. Supported by Sennrich et al. (2016), Wei & Zou (2019), Miller (1995).

Dataset	Source	Pairs	Avg Length	Purpose
Semantic-KG (Codex)	KG-perturbed	92	~69 words	Original baseline
MRPC	Human-annotated news	400	~20 words	Short-text lower bound
PAWS-Wiki	Word-scrambled Wikipedia	400	~20 words	Adversarial structural probe
Semantic-KG Combined	KG-perturbed (4 domains)	400	~69 words	Extended KG coverage
Wikipedia Entity-Swap	NLP-perturbed Wikipedia	399	~126 words	Anti-circularity probe

Semantic-KG (Codex) Results (paragraph-level, KG-perturbed)

Method	F1	AUC
AA-KEA	0.980	0.976
Sentence-T5	0.860	0.908
ROUGE-1	0.843	0.887

- AA-KEA dominates on intended paragraph-level domain

MRPC Results (Short Text)

Method	F1	AUC
Sentence-T5	0.765	0.816
BERTScore	0.758	0.816
ROUGE-1	0.745	0.784
AA-KEA	0.682	0.658

- AA-KEA ranks last; Root cause: at ~20 words per pair, KG extraction yields only 2–3 triples — too sparse for reliable graph comparison. Additionally, MRPC contains entity name variations (e.g., "Coke" vs "Coca-Cola") that the exact-string graph matcher treats as different nodes.

PAWS-Wiki Results (Adversarial)

Method	F1	AUC
AA-KEA	0.764	0.731
ROUGE-L	0.735	0.807
BERTScore	0.691	0.702
ROUGE-1	0.678	0.490

- AA-KEA ranks first. ROUGE-1 collapses to near-random (AUC=0.490) because PAWS-Wiki negatives are constructed by swapping subject/object positions — same words, inverted meaning. AA-KEA's KG structure detects the inversion; ROUGE only counts words.

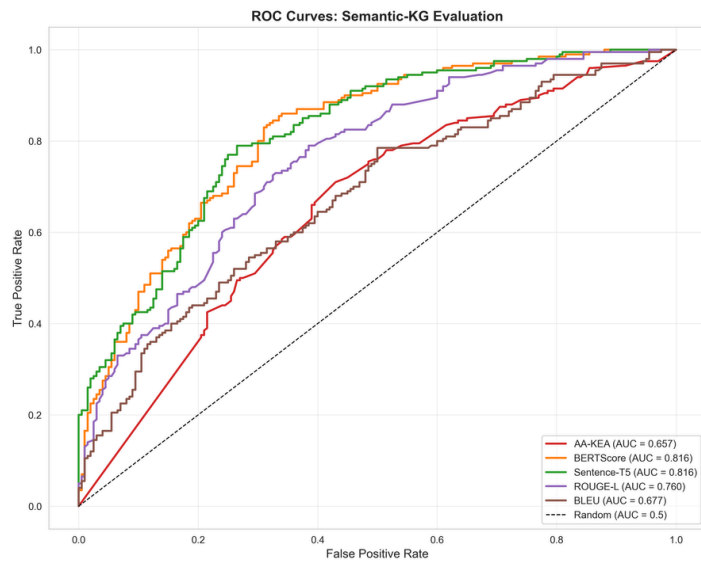
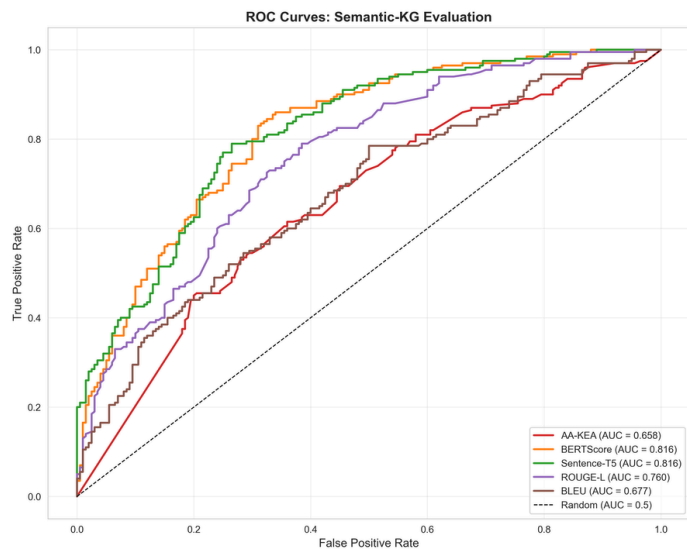
MRPC Method Variants Analysis

(three enhanced versions evaluated)

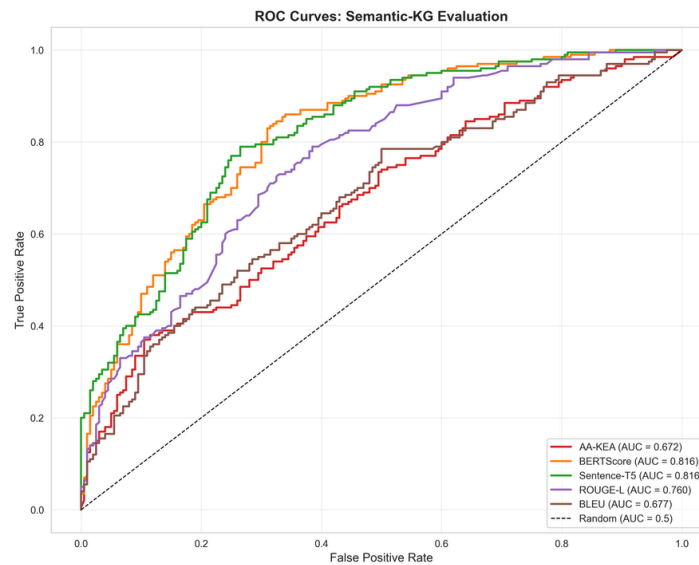
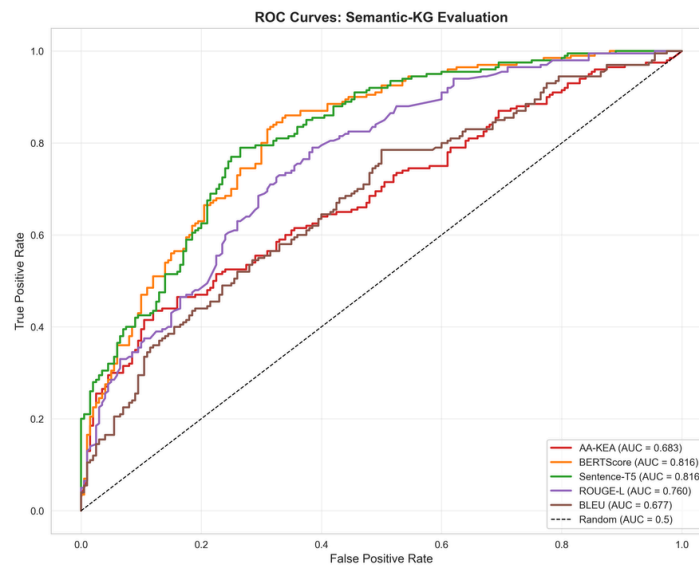
Variant	F1	AUC
Original AA-KEA	0.682	0.658
WL Similarity	0.681	0.683
WL Similarity Enhanced	0.682	0.672
KEA Enhanced	0.682	0.657

- All variants cluster tightly in F1 (~ 0.682), confirming the bottleneck is the dataset characteristics (short text, entity name variation), not the specific graph similarity method. WL Similarity shows marginally better AUC due to more continuous score distribution, which also explains the more convex ROC curve compared to the discrete-scoring original AA-KEA.

KEA BASED



WL BASED



Key Findings

1. Text richness is primary driver

: AA-KEA F1 drops from 0.980 (69-word paragraphs) to 0.682 (20-word sentences). Short text produces sparse KGs where a single mismatched node dominates the similarity score.

2. Structural advantage on adversarial data

: On PAWS-Wiki, AA-KEA ranks first while ROUGE-1 is near-random. This shows AA-KEA is not uniformly weak on short text — it is specifically weak when entity names vary across paraphrase pairs, and specifically strong when meaning changes through structural reordering that word-overlap methods cannot detect

3. Score granularity limits ROC smoothness on sparse KGs

: Original AA-KEA produces discrete scores (multiples of $1/n_{\text{triples}}$) on sparse KGs, causing a linear ROC curve start — both true and false positives share the same score buckets. WL and enhanced variants show improved score continuity.

Pending Work

- Semantic-KG Combined results awaited
- Wikipedia Entity-Swap final 40 pairs processing
- Cross-dataset plots and anti-circularity analysis pending

Limitations

- KG extraction uses Mistral-7B (not SOTA)
- Represents lower bound of AA-KEA capability
- Stronger LLM with entity linking and coreference resolution could improve short-text performance